Notes # 8

Uniqueness

Suppose there exists two solutions in a source free lossy region

$$\nabla\times\mathbf{E}_1 = -j\omega\mu\mathbf{H}_1 \qquad (1a) \qquad\qquad \nabla\times\mathbf{E}_2 = -j\omega\mu\mathbf{H}_2 \qquad (1b)$$

$$\nabla\times\mathbf{H}_1 = \mathbf{J}_1 + j\omega\varepsilon\mathbf{E}_1 \qquad (2a) \qquad\qquad \nabla\times\mathbf{H}_2 = \mathbf{J}_2 + j\omega\varepsilon\mathbf{E}_2 \qquad (2b)$$

$$\nabla\cdot\mathbf{E}_1 = \frac{\rho}{\varepsilon} \qquad (3a) \qquad\qquad \nabla\cdot\mathbf{E}_2 = \frac{\rho}{\varepsilon} \qquad (3b)$$

$$\nabla\cdot\mathbf{H}_1 = 0 \qquad (4a) \qquad\qquad \nabla\cdot\mathbf{H}_2 = 0 \qquad (4b)$$

which satisfy Maxwell's equations, the boundary conditions, and the radiation conditions.

If the two solutions are subtracted from one another the result must satisfy the source free Maxwell equations. To show this, let $\mathbf{J}_{\substack{1\\2}} = \sigma\mathbf{E}_{\substack{1\\2}}$ and subtract the above sets of equations 1b-1a, 2b-2a, etc.

$$\nabla\times(\mathbf{E}_2 - \mathbf{E}_1) = -j\omega\mu(\mathbf{H}_2 - \mathbf{H}_1), \qquad (5)$$

$$\nabla\times(\mathbf{H}_2 - \mathbf{H}_1) = (\sigma + j\omega\varepsilon)(\mathbf{E}_2 - \mathbf{E}_1) \quad (6) \text{ etc.}$$

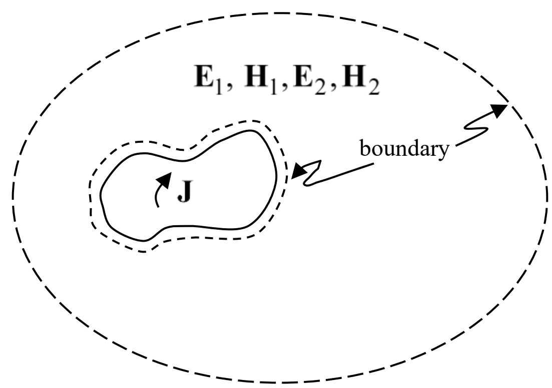
Then define $\mathbf{E}_2 - \mathbf{E}_1 = \mathbf{E}$ and $\mathbf{H}_2 - \mathbf{H}_1 = \mathbf{H}$ (7)
which gives:

$$\nabla\times\mathbf{E} = -j\omega\mu\mathbf{H} \qquad (8)$$

$$\nabla\times\mathbf{H} = (\sigma + j\omega\varepsilon)\mathbf{E} \qquad (9)$$

$$\nabla\cdot\mathbf{E} = 0 \qquad (10)$$

$$\nabla\cdot\mathbf{H} = 0 \qquad (11)$$



Take the dot product of (8) with $\mathbf{H}^*$ and the conjugate of (9) with $\mathbf{E}$ and subtract. This gives

$$\nabla\cdot\left(\mathbf{E}\times\mathbf{H}^*\right) = -\sigma E^2 + j\omega\varepsilon E^2 - j\omega\mu H^2 \quad (12)$$

where

$$\nabla\cdot\left(\mathbf{E}\times\mathbf{H}^*\right) = \mathbf{H}^*\cdot\nabla\times\mathbf{E} - \mathbf{E}\cdot\nabla\times\mathbf{H}^* \text{ has been used}$$

Integrate over a volume V and apply the divergence theorem to the $L.H.S.$ of the result

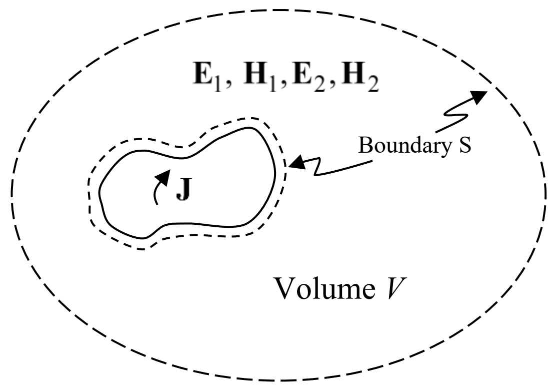
$$\oint_S \left(\mathbf{E}\times\mathbf{H}^*\right)\cdot d\mathbf{s} = -\int_V \sigma E^2 d\upsilon + j\omega\int_V \left(\varepsilon E^2 - \mu H^2\right) dv \quad (13)$$

$$\oint_S \left(\mathbf{E}\times\mathbf{H}^*\right)\cdot d\mathbf{s} = \int_V \sigma E^2 dv + j\omega\int_V \left(\varepsilon E^2 - \mu H^2\right) dv \quad (13)$$

Note: The two solutions $E_{\substack{1\\2}}$ and $H_{\substack{1\\2}}$ satisfy the boundary conditions on the boundary S so they must be equal on S. Therefore on S

$\mathbf{E} = \mathbf{E}_2 - \mathbf{E}_1 = 0$ and $\mathbf{H} = \mathbf{H}_2 - \mathbf{H}_1 = 0$ so,

$\oint_S \left(\mathbf{E}\times\mathbf{H}^*\right)\cdot d\mathbf{s} = 0$.



Since the $L.H.S.$ of (13) is zero, equate real and imaginary parts of the $R.H.S.$ to zero

- Real part:

  $$\int \sigma E^2 dv = 0 \Rightarrow E^2 = 0 \text{ or } E = 0 \text{ in } V$$

- Imag part: Since $E^2 = 0$ from above then

  $$j\omega\int_V \varepsilon E^2 dv = 0$$

  and therefore

  $$j\omega\int_V \mu H^2 dv = 0 \Rightarrow H^2 = 0 \text{ or } H = 0 \text{ in } V$$

Since $E = |\mathbf{E}|$ and $H = |\mathbf{H}|$ then

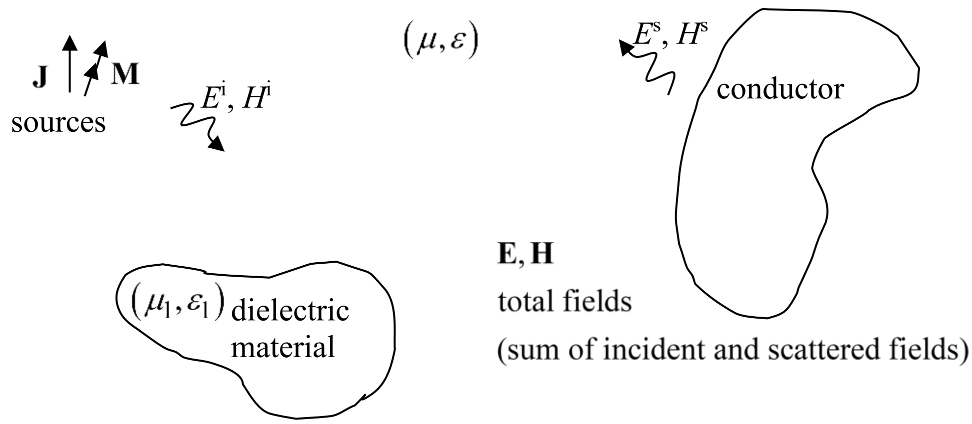
$\mathbf{E}_2 - \mathbf{E}_1 = \mathbf{E} = 0$ and $\mathbf{H}_2 - \mathbf{H}_1 = \mathbf{H} = 0$ throughout the volume V

$\Rightarrow \mathbf{E}_2 = \mathbf{E}_1$ and $\mathbf{H}_2 = \mathbf{H}_1$

$\therefore$ If the fields in a region satisfy Maxwell's equations and the boundary conditions then they are unique – That is, the two solutions must be the same (are unique) in the volume contained by the boundary.

Equivalence Principle

Suppose the sources radiate in the presence of some objects



Then we cannot use

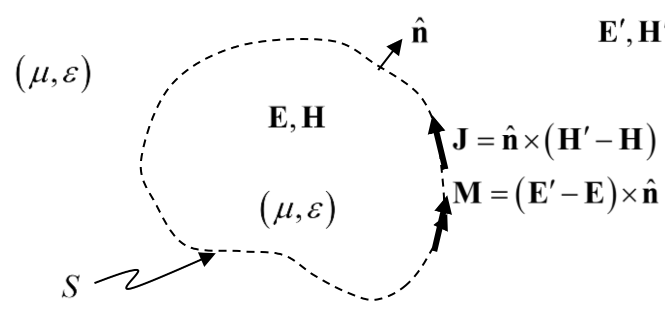
$$\mathbf{A} = \frac{\mu}{4\pi}\int_{v'} \mathbf{J}\frac{e^{-jkR}}{R}dv' \qquad (1a)$$

$$\mathbf{F} = \frac{\varepsilon}{4\pi}\int_{v'} \mathbf{M}\frac{e^{-jkR}}{R}dv' \qquad (1b)$$

to find the fields in the problem because this particular vector potential $\mathbf{A}$ does not satisfy the boundary conditions created by the presence of the 2 bodies.

Construction of the equivalent problem:

Suppose we want to find fields inside a dashed region shown above. If we had been able to solve the original problem, then we would know the fields on the dashed surface and therefore we could construct the interior equivalent problem as follows:



Note:
(i) The medium (μ, ε) is now the same everywhere so (1a) and (1b) can be used.

(ii) $\mathbf{E}$ and $\mathbf{H}$ inside the dashed region remain the same as in the original problem.

(iii) Note that $\mathbf{E}'$ and $\mathbf{H}'$ are completely arbitrary. This is so because we only want to maintain the field inside S. If we change $\mathbf{E}'$ and $\mathbf{H}'$, $\mathbf{J}$ and $\mathbf{M}$ will change so as to maintain the interior $\mathbf{E}$ and $\mathbf{H}$.

$$\mathbf{J} = \hat{\mathbf{n}} \times (\mathbf{H}' - \mathbf{H})$$

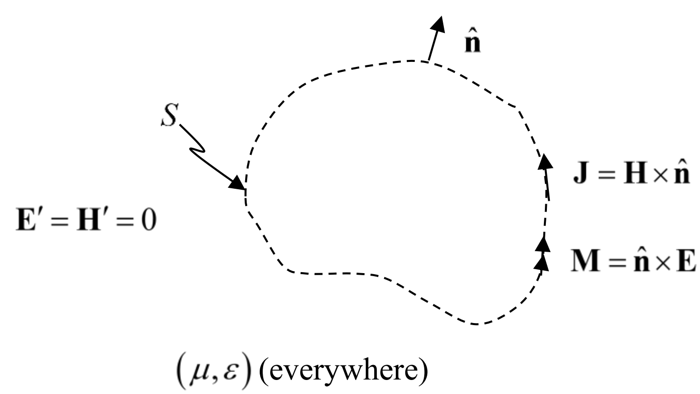
$$\mathbf{M} = (\mathbf{E}' - \mathbf{E}) \times \hat{\mathbf{n}}$$

$f(\mathbf{E}', \mathbf{H}', \mathbf{E}, \mathbf{H})$ arbitrary fixed

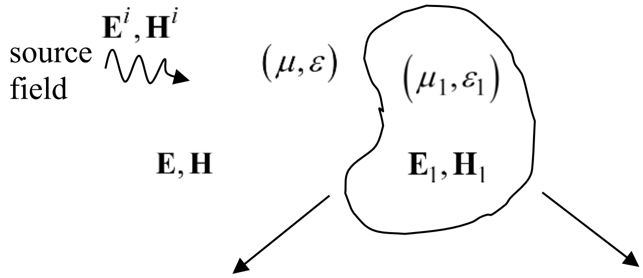
This means that $\mathbf{J}$ and $\mathbf{M}$ are not unique.

Since we can set $\mathbf{E}'$ and $\mathbf{H}'$ equal to anything, then in Equivalence Principle problems we always let $\mathbf{E}' = \mathbf{H}' = 0$

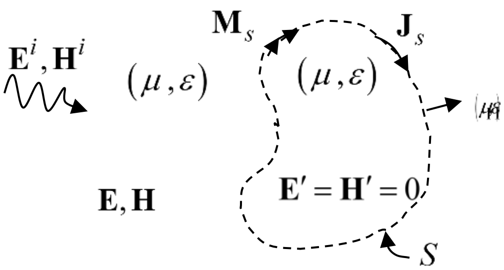
So if the fields in the original problem are known everywhere then the original problem (original sources, dielectric and conductor) can be replaced by

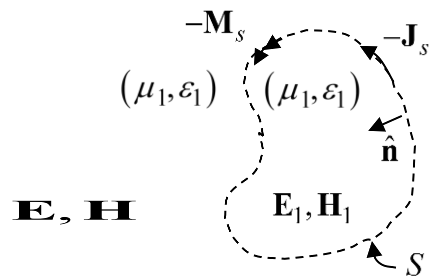


Example: Equivalence Principle applied to find the field scattered by a dielectric body



Exterior problem (see pg 4-11) Interior Problem





$$\mathbf{E} = \mathbf{E}^{scat} + \mathbf{E}^i$$

$$\mathbf{E} = -\frac{j\omega\mu}{k^2}\left(k^2 + \nabla\nabla\cdot\right)\int_S \mathbf{J}_s(\mathbf{r}')\frac{e^{-jk|\mathbf{r}-\mathbf{r}'|}}{4\pi|\mathbf{r}-\mathbf{r}'|}ds' - \nabla\times\int_S \mathbf{M}_s(\mathbf{r}')\frac{e^{-jk|\mathbf{r}-\mathbf{r}'|}}{4\pi|\mathbf{r}-\mathbf{r}'|}ds' + \mathbf{E}^i$$

$$\mathbf{H} = \mathbf{H}^{scat} + \mathbf{H}^i$$

$$\mathbf{H} = \nabla\times\int_S \mathbf{J}_s(\mathbf{r}')\frac{e^{-jk|\mathbf{r}-\mathbf{r}'|}}{4\pi|\mathbf{r}-\mathbf{r}'|}ds' - \frac{j\omega\varepsilon}{k^2}\left(k^2 + \nabla\nabla\cdot\right)\int_S \mathbf{M}_s(\mathbf{r}')\frac{e^{-jk|\mathbf{r}-\mathbf{r}'|}}{4\pi|\mathbf{r}-\mathbf{r}'|}ds' + \mathbf{H}^i$$

$$\mathbf{E}_1 = \frac{-j\omega\mu_1}{k_1^2}\left(k_1^2 + \nabla\nabla\cdot\right)\int_S \left(-\mathbf{J}_s(\mathbf{r}')\right)\frac{e^{-jk_1|\mathbf{r}-\mathbf{r}'|}}{4\pi|\mathbf{r}-\mathbf{r}'|}ds' - \nabla\times\int_S \left(-\mathbf{M}_s(\mathbf{r}')\right)\frac{e^{-jk_1|\mathbf{r}-\mathbf{r}'|}}{4\pi|\mathbf{r}-\mathbf{r}'|}ds'$$

$$\mathbf{H}_1 = \nabla\times\int_S \left(-\mathbf{J}_s(\mathbf{r}')\right)\frac{e^{-jk_1|\mathbf{r}-\mathbf{r}'|}}{4\pi|\mathbf{r}-\mathbf{r}'|}ds' - \frac{j\omega\varepsilon_1}{k_1^2}\left(k_1^2 + \nabla\nabla\cdot\right)\int_S \left(-\mathbf{M}_s(\mathbf{r}')\right)\frac{e^{-jk_1|\mathbf{r}-\mathbf{r}'|}}{4\pi|\mathbf{r}-\mathbf{r}'|}ds'$$

Note: The surface currents $\mathbf{J}_s$ (inside) $= -\mathbf{J}_s$ (outside) and $\mathbf{M}_s$ (inside) $= -\mathbf{M}_s$ (outside) because the tangential components of $\mathbf{E}$ and $\mathbf{H}$ are continuous across the boundary and the $\hat{\mathbf{n}}$'s are in opposite directions. $\hat{\mathbf{n}}_{out}\times\left(\mathbf{H}^{outside} - \mathbf{H}^{inside}\right) = 0 \Rightarrow \hat{\mathbf{n}}_{out}\times\mathbf{H}^{outside} = \hat{\mathbf{n}}_{out}\times\mathbf{H}^{inside}$ but $-\hat{\mathbf{n}}_{out}\times\mathbf{H}^{inside} = \hat{\mathbf{n}}_{in}\times\mathbf{H}^{inside}$ requires that $\mathbf{J}_s$ (inside) $= -\mathbf{J}_s$ (outside).

Continuity of electric field across surface:

$$\hat{\mathbf{n}}\times\left(\mathbf{E}-\mathbf{E}_1\right)\underset{\mathbf{r}=\mathbf{s}}{=}0$$

$$\frac{j\omega\mu}{k^2}\hat{\mathbf{n}}\times\left(k^2+\nabla\nabla\cdot\right)\int_S\frac{\mathbf{J}_s\left(\mathbf{s}'\right)e^{-jk|\mathbf{s}-\mathbf{s}'|}}{4\pi|\mathbf{s}-\mathbf{s}'|}ds'$$

$$+\frac{j\omega\mu_1}{k_1^2}\hat{\mathbf{n}}\times\left(k_1^2+\nabla\nabla\cdot\right)\int_S\frac{\mathbf{J}_s\left(\mathbf{s}'\right)e^{-jk_1|\mathbf{s}-\mathbf{s}'|}}{4\pi|\mathbf{s}-\mathbf{s}'|}ds'$$

$$+\hat{\mathbf{n}}\times\nabla\times\int_S\frac{\mathbf{M}_s(\mathbf{s}')e^{-jk|\mathbf{s}-\mathbf{s}'|}}{4\pi|\mathbf{s}-\mathbf{s}'|}ds'+\hat{\mathbf{n}}\times\nabla\times\int_S\frac{\mathbf{M}_s(\mathbf{s}')e^{-jk_1|\mathbf{s}-\mathbf{s}'|}}{4\pi|\mathbf{s}-\mathbf{s}'|}ds'$$

$$=\hat{\mathbf{n}}\times\mathbf{E}^i \qquad \text{(a)}$$

and continuity of magnetic field across surface:

$$\hat{\mathbf{n}}\times\left(\mathbf{H}-\mathbf{H}_1\right)=0$$

$$\hat{\mathbf{n}}\times\nabla\times\int_S\frac{\mathbf{J}_s\left(\mathbf{s}'\right)e^{-jk|\mathbf{s}-\mathbf{s}'|}}{4\pi|\mathbf{s}-\mathbf{s}'|}ds'+\hat{\mathbf{n}}\times\nabla\times\int_S\frac{\mathbf{J}_s\left(\mathbf{s}'\right)e^{-jk_1|\mathbf{s}-\mathbf{s}'|}}{4\pi|\mathbf{s}-\mathbf{s}'|}ds'$$

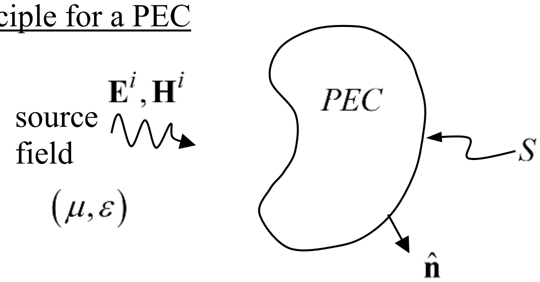
$$\frac{-j\omega\varepsilon}{k^2}\hat{\mathbf{n}}\times\left(k^2+\nabla\nabla\cdot\right)\int_S\frac{\mathbf{M}_s\left(\mathbf{s}'\right)e^{-jk|\mathbf{s}-\mathbf{s}'|}}{4\pi|\mathbf{s}-\mathbf{s}'|}ds'$$

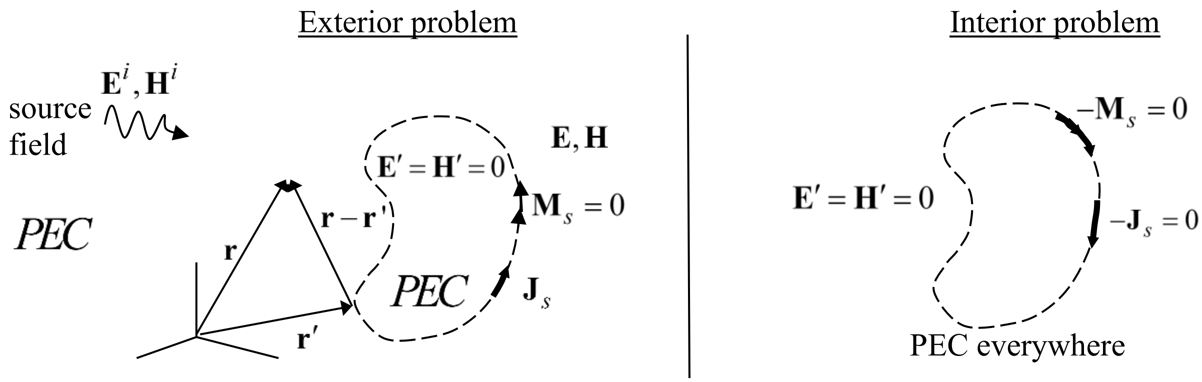
$$\frac{-j\omega\varepsilon_1}{k_1^2}\hat{\mathbf{n}}\times\left(k_1^2+\nabla\nabla\cdot\right)\int_S\frac{\mathbf{M}_s(\mathbf{s}')e^{-jk_1|\mathbf{s}-\mathbf{s}'|}}{4\pi|\mathbf{s}-\mathbf{s}'|}ds'$$

$$=-\hat{\mathbf{n}}\times\mathbf{H}^i\left(=-\hat{\mathbf{n}}\times\frac{\mathbf{k}\times\mathbf{E}^i}{\eta}\text{(if a plane wave)}\right) \quad \text{(b)}$$

$\therefore$ (a) & (b) are 2 eqns with 2 unknowns. There is no analytical solution. The method of moments (MoM) numerical method is used to obtain an approximate solution for $\mathbf{J}_s$ and $\mathbf{M}_s$.

Example: Equivalence Principle for a PEC





The electric field due to the source $(\mathbf{J}_s)$ and the incident field $(\mathbf{E}^i)$ make up the exterior field

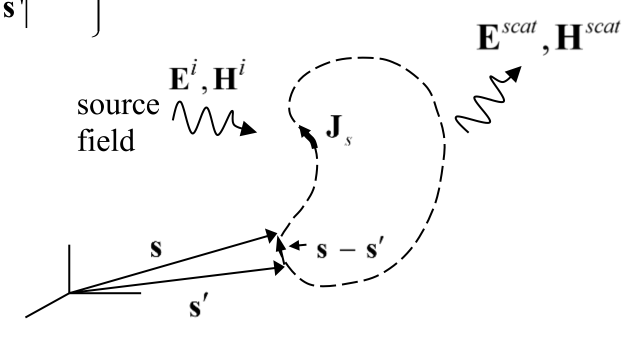
$$\mathbf{E} = \mathbf{E}^{scat} + \mathbf{E}^i = -\frac{j\omega}{k^2}\left(k^2\mathbf{A} + \nabla\nabla\cdot\mathbf{A}\right) + \mathbf{E}^i$$

Since
$$\mathbf{n} \times \left(\mathbf{E} - \cancelto{0}{\mathbf{E}_1}\right)_{\mathbf{r}=\mathbf{s}} = 0$$
$$\hat{\mathbf{n}} \times \left(\mathbf{E}^{scat} + \mathbf{E}^i\right) = 0$$
$$\hat{\mathbf{n}} \times \mathbf{E}^{scat} = -\hat{\mathbf{n}} \times \mathbf{E}^i$$

The complete integral equation is the <u>Electric field integral equation:</u>

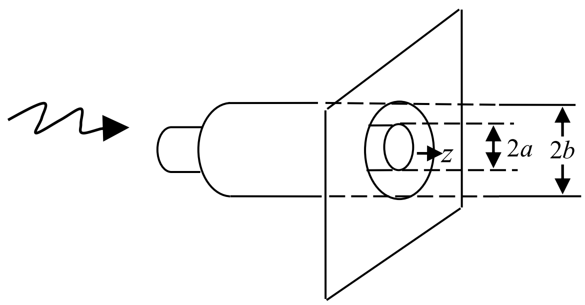
$$\frac{-j\omega\mu}{4\pi k^2}\hat{\mathbf{n}} \times \left\{\left(k^2 + \nabla\nabla\cdot\right)\iint_s \mathbf{J}_s(\mathbf{s}')\frac{e^{-jk|\mathbf{s}-\mathbf{s}'|}}{|\mathbf{s}-\mathbf{s}'|}ds'\right\} = -\hat{\mathbf{n}} \times \mathbf{E}^i(s)$$

only unknown



Due to the Equivalence Principle formulation and B.C.'s, $\mathbf{J}_s$ is the actual current on the surface of the conductor.

Sect 3.6 – Approximate solution for a coaxial line opening into a conducting plane $(b << \lambda)$



<u>Interior Field</u>

$$\mathbf{E}^{\text{int}} = \mathbf{E}^{+} + \mathbf{E}^{-}$$

$$\doteq -\frac{V_0}{\rho \ln(b/a)} e^{-jkz} \hat{\boldsymbol{\rho}} - \frac{V_0}{\rho \ln(b/a)} e^{jkz} \hat{\boldsymbol{\rho}}$$

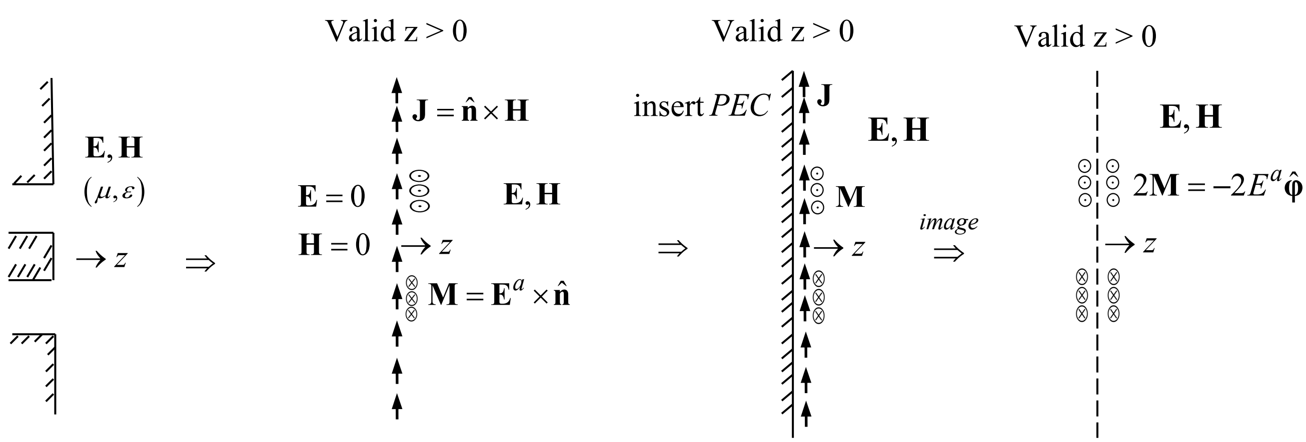
(assuming only TEM mode and neglecting higher order modes)

incident TEM mode

reflection coefficient at an open circuit $\simeq +1$

- In the aperture $(z=0)$

$$\mathbf{E}^{\text{int}} = \mathbf{E}^{a} \simeq -\frac{V}{\rho \ln(b/a)} \hat{\boldsymbol{\rho}} \qquad \text{where } V = 2V_0 \text{ at } z = 0$$



$$\mathbf{M} = \mathbf{E}^{a} \times \hat{\mathbf{n}} = E^{a} \hat{\boldsymbol{\rho}} \times \hat{\mathbf{z}} = -E^{a} \hat{\boldsymbol{\varphi}} = \frac{V}{\rho \ln b/a} \hat{\boldsymbol{\varphi}}$$

$$\mathbf{E} = -\frac{1}{\varepsilon} \nabla \times \mathbf{F} = -\nabla \times \frac{1}{4\pi} \int_{\rho'=a}^{b} \int_{\phi'=0}^{2\pi} 2\mathbf{M}(\rho') \frac{e^{-jkR}}{R} \rho' d\rho' d\phi'$$

$$\mathbf{E} = -\nabla \times \frac{1}{4\pi} \int_{\rho'=a}^{b} \int_{\phi'=0}^{2\pi} \frac{2V}{\cancel{\rho'} \ln b/a} \hat{\boldsymbol{\varphi}}' \frac{e^{-jkR}}{R} \cancel{\rho'} d\phi' d\rho'$$

$$\begin{aligned} \hat{\boldsymbol{\varphi}}' &= (\hat{\boldsymbol{\varphi}}' \cdot \hat{\mathbf{r}})\hat{\mathbf{r}} + (\hat{\boldsymbol{\varphi}}' \cdot \hat{\boldsymbol{\theta}})\hat{\boldsymbol{\theta}} + (\hat{\boldsymbol{\varphi}}' \cdot \hat{\boldsymbol{\varphi}})\hat{\boldsymbol{\varphi}} \\ &= \sin\theta \sin(\phi - \phi')\hat{\mathbf{r}} + \cos\theta \sin(\phi - \phi')\hat{\boldsymbol{\theta}} + \cos(\phi - \phi')\hat{\boldsymbol{\varphi}} \end{aligned}$$

(from notes #5)

The magnetic current is independent of ϕ so the field will be ϕ independent $\Rightarrow$ choose $\phi = 0$

$\hat{\boldsymbol{\varphi}}' = -\sin\theta \sin\phi' \hat{\mathbf{r}} - \cos\theta \sin\phi' \hat{\boldsymbol{\theta}} + \cos\phi' \hat{\boldsymbol{\varphi}}$

$$\frac{e^{-jkR}}{R} = \frac{e^{-jk(r - \rho' \cos\phi' \sin\theta)}}{r}$$

$$\mathbf{E} = -\frac{2V}{4\pi \ln(b/a)} \nabla \times \frac{e^{-jkr}}{r} \int_{\rho'=a}^{b} \left\{ \int_{\phi'=0}^{2\pi} e^{jk\rho' \cos\phi' \sin\theta} \times \left[\underbrace{-\sin\theta \sin\phi' \hat{\mathbf{r}}}_{I_r} \underbrace{- \cos\theta \sin\phi' \hat{\boldsymbol{\theta}}}_{I_\theta} + \underbrace{\cos\phi' \hat{\boldsymbol{\varphi}}}_{I_\phi} \right] d\phi' \right\} d\rho'$$

$$I_\phi = \int_{\phi'=0}^{2\pi} \cos\phi' e^{jk\rho' \cos\phi' \sin\theta} d\phi' = j2\pi J_1(k\rho' \sin\theta)$$ from (9.1.21) A&S

$$\doteq j2\pi \frac{k\rho'}{2} \sin\theta$$ (when $\rho' << \lambda$) from (9.1.7) A&S

$$= j\pi k\rho' \sin\theta$$

$$I_r = -\sin\theta \int_{\phi'=0}^{2\pi} \sin\phi' e^{jk\rho' \cos\phi' \sin\theta} d\phi' = -\sin\theta \frac{2\pi^{1/2}}{\left(\frac{k\rho' \sin\theta}{2}\right)^{1/2}} J_{1/2}(k\rho' \sin\theta) \text{ (9.1.20)A\&S}$$

$$= -\sin\theta \frac{2\pi^{1/2}}{\left(\frac{k\rho' \sin\theta}{2}\right)^{1/2}} \left[\left(\frac{2k\rho' \sin\theta}{\pi} \right)^{1/2} j_0(k\rho' \sin\theta) \right]$$

$$= -4\sin\theta\, j_0(k\rho' \sin\theta)$$

$$= -4\sin\theta \underbrace{\frac{\sin(k\rho' \sin\theta)}{k\rho' \sin\theta}}_{\simeq 1\ (\rho' << \lambda)}$$

(10.1.11) A&S

$$= -4\sin\theta$$

$$I_\theta = -\cos\theta \int_{\phi'=0}^{2\pi} \sin\phi' e^{jk\rho'\cos\phi'\sin\theta} d\phi' = -4\cos\theta$$

$$\mathbf{E} = -\frac{V}{2\pi\ln\left(b/a\right)} \nabla\times\frac{e^{-jkr}}{r}\left\{ j\pi k\sin\theta \int_{\rho'=a}^{b} \rho' d\rho' \hat{\boldsymbol{\varphi}} \right.$$

$$\left. -4\left(\sin\theta\hat{\mathbf{r}} + \cos\theta\hat{\boldsymbol{\theta}}\right) \int_{\rho'=a}^{b} d\rho' \right\}$$

$$= -\frac{V}{2\pi\ln\left(b/a\right)} \nabla\times\frac{e^{-jkr}}{r}\left[\frac{j\pi k\sin\theta}{2}\left(b^2-a^2\right)\hat{\boldsymbol{\varphi}} \right.$$

$$\left. -4\left(b-a\right)\left(\sin\theta\hat{\mathbf{r}} + \cos\theta\hat{\boldsymbol{\theta}}\right)\right]$$

$$= -\frac{V}{2\pi\ln\left(b/a\right)} \left\{ \frac{j\pi k}{2}\left(b^2-a^2\right)\left[\frac{e^{-jkr}}{r}\left(\frac{1}{r\sin\theta}\frac{\partial}{\partial\theta}\sin^2\theta\right)\hat{\mathbf{r}} \right.\right.$$

$$\left. -\frac{1}{r}\sin\theta\frac{\partial}{\partial r}e^{-jkr}\hat{\boldsymbol{\theta}}\right]$$

$$+4\left(b-a\right)\left[\frac{e^{-jkr}}{r}\left(\frac{1}{r}\frac{\partial}{\partial\theta}\sin\theta\right)\right]\hat{\boldsymbol{\varphi}}$$

$$\left. -4\left(b-a\right)\left[\cos\theta\left(\frac{1}{r}\frac{\partial}{\partial r}r\frac{e^{-jkr}}{r}\right)\right]\hat{\boldsymbol{\varphi}}\right\}$$

$$= -j\frac{V}{4\ln b/a}k\left(b^2-a^2\right)\left[-\frac{1}{r}\sin\theta\left(-jk\right)e^{-jkr}\right]\hat{\boldsymbol{\theta}} - \frac{V}{2\pi\ln\left(b/a\right)}$$

$$\mathbf{E} = \frac{Vk^2}{4\ln b/a}\left(b^2-a^2\right)\frac{e^{-jkr}}{r}\sin\theta\hat{\boldsymbol{\theta}} - j\frac{2Vk}{\pi\ln b/a}\left(b-a\right)\frac{e^{-jkr}}{r}\cos\theta\hat{\boldsymbol{\varphi}}\,?$$

$$H_\phi = \frac{Vk^2}{4\eta\ln b/a}\left(b^2-a^2\right)\frac{e^{-jkr}}{r}\sin\theta \rightarrow \text{ with } \frac{k^2}{4\eta} \rightarrow \frac{\omega\varepsilon\pi}{2\lambda}$$